

# Explainable Multi-Threshold Machine Learning Analysis of Depression Among Bangladeshi Adolescents: An Empirical Study

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**Abstract**—Depression in adolescents is a growing mental health problem in developing countries like Bangladesh. Machine learning-based depression risk prediction among Bangladeshi adolescents remains underexplored. Using ML algorithms, including classical models, ensemble methods, deep learning, and meta-learners we analyzed an open-access dataset of 1,919 Bangladeshi college-going adolescents. Depression were measured using the PHQ-9 scale. For classification, we used two clinically established thresholds: scores of 10 or higher indicating moderate-to-severe risk, and scores of 15 or higher indicating high risk of depression. At the moderate threshold (PHQ-9  $\geq 10$ ), Cat\_GB\_Stacking (CatBoost + GradientBoosting) achieved an accuracy of 80.7% and AUC of 0.757. At the severe threshold (PHQ-9  $\geq 15$ ), the Ada\_Log\_Stacking (AdaBoost + Logistic Regression) attained 70.3% accuracy with AUC 0.762. Friedman statistical testing confirmed significant performance differences among all 35 models at both the moderate threshold ( $\chi^2 = 290.85, p < 0.001$ ) and the severe threshold ( $\chi^2 = 203.80, p < 0.001$ ), demonstrating that model performance varies significantly regardless of the depression severity level. For all the models, 10-fold stratified cross validation was applied to avoid overfitting. Family unrest emerged as the most influential predictor across both thresholds, achieving feature importance scores of 0.618 and 0.556 respectively. The overall analysis provides a validated framework for identifying adolescents at risk of depression and highlights the critical role of family environment in adolescent mental health.

**Index Terms**—Adolescent depression; Machine learning; Deep learning; Ensemble methods; Friedman test; PHQ-9; Family unrest; Bangladesh; Mental health screening

## I. INTRODUCTION

Adolescence is the transitional phase between childhood and adulthood, defined as the age range of 10 to 19 years. During the adolescent phase, individuals undergo quick physical and psychosocial changes. The World Health Organization (WHO) defines depression as a mood and behavioral disorder characterized by persistent alterations in emotional state or pleasure in daily activities. According to the WHO, approximately 20% of adolescents worldwide experience mental health disorders [1]. Adolescent are at greater risks of depression which is influenced by academic performance, family environment, parenting practices, and psychosocial stressors [2]. Prior studies of depression have identified factors such as gender, sleep quality, and age as significant predictors of depressive symptoms [3], [4]. The dataset we used in this

study was originally published and analyzed using SPSS-based statistical methods [5], the present study extends prior work by systematically evaluating and comparing established ML models within a multi-threshold PHQ-9 framework in the context of Bangladeshi adolescent. Several ML-based depression studies conducted in Bangladesh have primarily focused on university-going students rather than adolescents [6]. Many of these studies relied on single-threshold binary classification approaches using algorithms such as Random Forest and Support Vector Machines [7]. The main contributions of this work are summarized as follows:

- 1) This study focuses on underrepresented Bangladeshi adolescents.
- 2) Comparison of different predictive models, including classical ML, ensemble methods, deep learning, and stacking meta-learners.
- 3) Multi-threshold analysis (PHQ-9  $\geq 10$  and PHQ-9  $\geq 15$ ) with non-parametric statistical validation using Friedman and Nemenyi post-hoc tests.
- 4) SHAP and LIME analysis to interpret feature contributions to depression prediction outcomes.

## II. LITERATURE REVIEW

Many Bangladeshi studies have applied ML techniques to predict mental health disorders. Nayan et al. [8] analyzed multiple ensemble models to predict depression among university students using sociodemographic variables. Similarly, Moon et al. [3] explored depression prediction among employees using factor analysis and ML algorithms, highlighting the impact of income and occupation. Masud et al. [6] predicted depression using multiple ML and deep learning models and applied SHAP to interpret predictions. Choudhury et al. [7] analyzed ML models for early suicide prevention. However, these studies were limited to single-threshold binary classification and mainly focused on university populations, limiting generalizability. Other Bangladeshi studies have used survey-based ML approaches in regional samples [9], focusing on local risk patterns but not incorporating multi-threshold analyses. Siddik et al. [5], which provided the current dataset, relied on traditional statistics, identifying family unrest as a key risk factor without ML. Overall, existing research shows three gaps: (i) focus on university populations rather than broader adolescents,

(ii) predictive modeling limited to single-threshold outcomes, and (iii) lack of systematic model comparison with statistical validation. The present study addresses these gaps using multi-threshold analysis with 35 ML models, Friedman testing, and comprehensive feature importance analysis.

Table I shows the comparative analysis of the literature review.

### III. MATERIALS AND METHODS

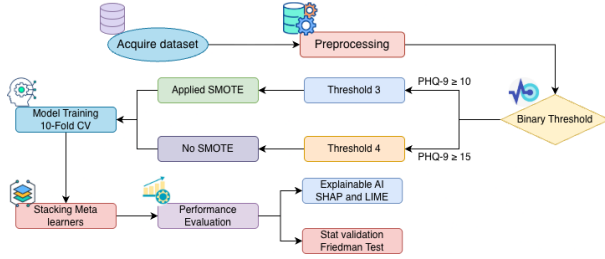


Fig. 1: Overview of the proposed methodological framework.

#### Dataset Description

This study utilized an open-access dataset originally collected by Siddik et al. [5]. There were socio-demographic variables (age, gender, religion, and family income), behavioral factors (social media usage, physical exercise, and smoking), relationship-related variables (pre-marital relationship and marital status), and psychosocial risk factors (experiencing blackmail, family unrest, chronic illness, and recovered from COVID-19) in the dataset. Depression was measured using the Patient Health Questionnaire-9 (PHQ-9). PHQ-9 scores were categorized into five severity levels: 1. minimal (0-4), 2. mild (5-9), 3. moderate (10-14), 4. moderately severe (15-19), and 5. severe (20-27). The severity levels 3 and 4 were used as threshold for classification.

#### Data Preprocessing

Ordinal variables, including grade point average, family income, and study duration, were encoded using ordinal encoding. All Yes/No variables were encoded as 0/1. Other nominal categorical variables were one-hot encoded. One missing value in age column were imputed using mean imputation. Feature scaling was applied using StandardScaler. The original multiclass (1-5) depression labels were transformed into two binary classification settings to examine threshold sensitivity:

- **Threshold 3 (Moderate-or-higher depression):** Participants with PHQ-9 scores  $\geq 10$  were classified as *risky* (positive class, 80.1%), while those with scores  $< 10$  were classified as *safe* (negative class, 19.9%).
- **Threshold 4 (Moderately severe-or-higher depression):** Participants with PHQ-9 scores  $\geq 15$  were classified as *risky* (positive class, 55.9%), whereas those with scores  $< 15$  were classified as *safe* (negative class, 44.1%). This stricter threshold captures individuals with severe depressive symptoms.

Due to class imbalance at Threshold 3 (80.1% positive class), SMOTE was applied only on the training folds during cross-validation to force the ML model to learn the minority class. As Threshold 4 had more balanced class distribution (55.9% vs 44.1%) it did not require SMOTE augmentation.

#### Model Training and Evaluation

A total of 35 supervised machine learning algorithms (including Logistic Regression, AdaBoost, Gradient Boosting, XGBoost, and CatBoost, ANN, CNN etc) and Stacking ensembles was implemented and trained using stratified 10-fold cross-validation. Performance was evaluated using accuracy, precision, recall, F1-score, and AUC. Stacking used out-of-fold base predictions to train a cross-validated Logistic Regression meta-learner, evaluated on the same folds to avoid overfitting. Feature importance was derived from the best-performing Logistic Regression models at both thresholds. LIME explained individual predictions by slightly changing input values and using a simple model to understand the prediction.

#### Statistical Validation: Friedman Test and Post-Hoc Analysis

Friedman test were used to check whether the performance differences between algorithms were statistically significant. The Friedman test was applied to all 35 individual models (stacking ensembles were excluded). If the Friedman test showed a significant result ( $p < 0.05$ ), then the Nemenyi post-hoc test were used to identify which algorithm pairs were significantly different, while controlling for multiple comparisons.

### IV. RESULTS

The dataset collected responses of 1,919 responses having mean age of 17.5 years, and 65.5% of them were female. Class imbalance varied in the two thresholds as expected. At Threshold 3 (PHQ-9  $\geq 10$ ), 80.1% participants were classified as at-risk and 19.9% as safe, indicating class imbalance. At Threshold 4 (PHQ-9  $\geq 15$ ), the distribution was 55.9% participants classified as at-risk and 44.1% as safe indicating class balance. Correlation analysis revealed moderate associations among predictor variables (Figure 2). Family unrest showed notable correlations with blackmail exposure ( $r = 0.35$ ) and internet addiction ( $r = 0.28$ ).

#### ML Model Performance

A total of 35 machine learning models were evaluated for both depression severity thresholds using 10-fold stratified cross-validation. The models included classical algorithms, ensemble methods, deep learning, and stacking meta-learners. Tables IV and V showing the best performing models Threshold 3 (PHQ-9  $\geq 10$ ) and Threshold 4 (PHQ-9  $\geq 15$ ).

1) *Moderate-or-Higher Depression Prediction (PHQ-9  $\geq 10$ ):* At Threshold 3, Cat\_GB\_Stacking achieved the highest accuracy (80.7%, F1 = 0.889, AUC = 0.757). Among individual models, CatBoost performed best (79.3% accuracy, AUC = 0.754), while deep learning showed moderate performance (ANN\_Medium: 73.5%, AUC = 0.731). The Dummy

TABLE I: Summary of prior studies on ML-based depression prediction in Bangladesh and related contexts

| Author & Year               | Country     | Target Population         | Methodology                                   | Key Findings                                                             | Limitations                                         |
|-----------------------------|-------------|---------------------------|-----------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------|
| Siddik et al. (2024) [5]    | Bangladesh  | 1,919 college adolescents | Chi-square, logistic regression               | High prevalence of depression; family unrest identified as key predictor | No machine learning used                            |
| Nayan et al. (2022) [8]     | Bangladesh  | 2,121 university students | Supervised ML (RF, SVM, LR, KNN, NB)          | Effective binary depression prediction during COVID-19                   | Non-representative sample; only university students |
| Moon et al. (2021) [3]      | Bangladesh  | Job sector employees      | Factor analysis and ML (RF, NB, KNN)          | Linked income and occupation to depression                               | Did not explore clinical severity thresholds        |
| Al Masud et al. (2025) [6]  | Bangladesh  | University students       | Deep learning and explainable AI (SHAP, LIME) | Improved model interpretability                                          | Relied on single-threshold classification           |
| Choudhury et al. (2019) [7] | Bangladesh  | University students       | Supervised ML (DT, SVM, RF, KNN)              | Early detection for suicide prevention                                   | Single-threshold binary classification              |
| Tabassum et al. (2025) [9]  | Bangladesh  | Teachers and students     | Survey-based supervised ML                    | Regional mental health insights (Sylhet)                                 | Limited multi-threshold analysis                    |
| Kim et al. (2024) [10]      | South Korea | 95 clinical adolescents   | Symptom cluster analysis                      | Identified clinically meaningful symptom clusters                        | Small, clinically restricted sample                 |

TABLE II: Description of Study Variables

| Variable           | Levels                    | Data Type             |
|--------------------|---------------------------|-----------------------|
| Age                | 15–18                     | Numeric               |
| Gender             | Male / Female             | Categorical – Nominal |
| Family_income      | <10,000 → >40,000         | Categorical – Nominal |
| social_media       | Yes / No (0/1)            | Categorical – Nominal |
| Religion           | Islam / Hinduism / Others | Categorical – Nominal |
| internet_addiction | Yes / No                  | Categorical – Nominal |
| exercise           | Yes / No                  | Categorical – Nominal |
| pre_marital        | Yes / No                  | Categorical – Nominal |
| practice_religion  | Yes / No                  | Categorical – Nominal |
| smoking            | Yes / No                  | Categorical – Nominal |
| blackmail          | Yes / No                  | Categorical – Nominal |
| family_unrest      | Yes / No                  | Categorical – Nominal |
| major_illness      | Yes / No                  | Categorical – Nominal |
| covid              | Yes / No                  | Categorical – Nominal |
| ssc                | SSC Grade Point Average   | Numeric               |
| mental_illness     | Yes / No                  | Categorical – Nominal |
| sex_experience     | Yes / No                  | Categorical – Nominal |
| marital_status     | Married / Unmarried       | Categorical – Nominal |

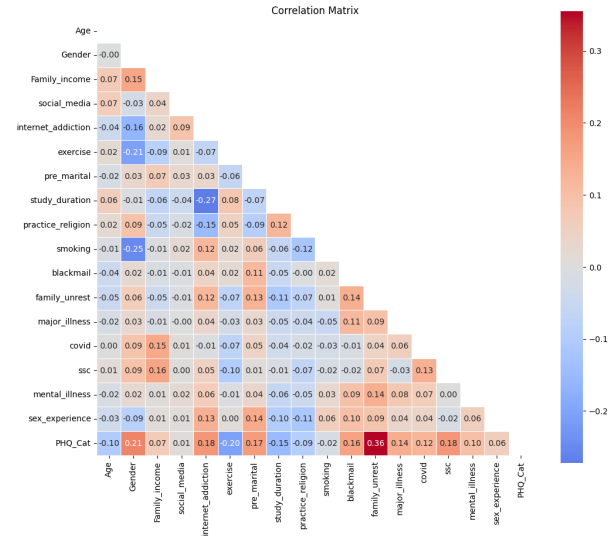


Fig. 2: Correlation matrix of predictor variables.

classifier reached only 19.9% accuracy (AUC = 0.500), confirming that trained models substantially outperformed random guessing. ROC curves (Figure 3) show AUC values between 0.72 and 0.76. In ablation analysis, CatBoost (79.26%) and Gradient Boosting (78.89%) were slightly inferior to the stacking ensemble (80.72%), demonstrating a +1.46% gain and improved generalization through combining complementary learners.

#### 2) Statistical Validation at Threshold 3: Friedman Test:

The Friedman test was conducted on the top 14 individual models to examine whether performance differences were statistically significant. Results are summarized in Table VI. At Threshold 3, the test produced  $\chi^2 = 290.85$  with  $p < 0.001$ , indicating highly significant differences among the 35 models. Subsequently, pairwise Nemenyi post-hoc comparisons were performed. Gradient boosting algorithms formed a statistically superior group compared to simpler tree-based models and basic ensembles (e.g., Bagging, AdaBoost). CatBoost, Gradient Boosting, and LightGBM showed no significant differences

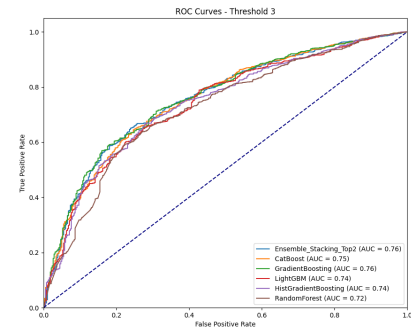
Fig. 3: ROC curves for top-performing models under PHQ-9  $\geq 10$  (Threshold 3).

TABLE III: Performance comparison of top models for different Thresholds.

TABLE IV: Threshold 3 (PHQ-9  $\geq 10$ )

| Model                          | Acc.          | F1            | Prec.         | Rec.          | AUC           |
|--------------------------------|---------------|---------------|---------------|---------------|---------------|
| <i>Ensemble Model</i>          |               |               |               |               |               |
| Cat_GB_Stacking                | <b>0.8072</b> | <b>0.8890</b> | <b>0.8429</b> | <b>0.9404</b> | 0.7567        |
| <i>Machine Learning Models</i> |               |               |               |               |               |
| CatBoost                       | 0.7926        | 0.8750        | 0.8401        | 0.9127        | 0.7537        |
| GradientBoosting               | 0.7889        | 0.8704        | 0.8363        | 0.9082        | <b>0.7597</b> |
| LightGBM                       | 0.7843        | 0.8701        | 0.8312        | 0.9137        | 0.7394        |
| HistGradientBoosting           | 0.7796        | 0.8664        | 0.8289        | 0.9082        | 0.7392        |
| RandomForest                   | 0.7775        | 0.8648        | 0.8303        | 0.9025        | 0.7211        |
| ExtraTrees                     | 0.7717        | 0.8600        | 0.8257        | 0.8982        | 0.6905        |
| XGBoost                        | 0.7671        | 0.8573        | 0.8230        | 0.8950        | 0.7180        |
| AdaBoost                       | 0.7342        | 0.8207        | 0.8051        | 0.8371        | 0.7647        |
| <i>Deep Learning Models</i>    |               |               |               |               |               |
| ANN_Medium                     | 0.7347        | 0.8259        | 0.8016        | 0.8520        | 0.7306        |
| ANN_Large                      | 0.7337        | 0.8259        | 0.7992        | 0.8546        | 0.7222        |
| CNN_1D                         | 0.7134        | 0.8100        | 0.7813        | 0.8408        | 0.7080        |
| <i>Baseline</i>                |               |               |               |               |               |
| Dummy                          | 0.1985        | 0.0000        | 0.0000        | 0.0000        | 0.5000        |

among themselves ( $p > 0.05$ ), but all significantly outperformed Decision Tree and ExtraTree classifiers ( $p < 0.001$ ). The MLP classifier performed comparably to most gradient boosting models and significantly better than the simplest tree classifiers.

TABLE VI: Friedman test results for statistical validation of model performance differences

| Threshold                      | $\chi^2$ | p-value   | Significance |
|--------------------------------|----------|-----------|--------------|
| Threshold 3 (PHQ-9 $\geq 10$ ) | 290.8530 | $< 0.001$ | Yes          |
| Threshold 4 (PHQ-9 $\geq 15$ ) | 203.8042 | $< 0.001$ | Yes          |

3) *Feature Importance Analysis at Threshold 3*: From L2-regularized Logistic Regression model, feature importance were evaluated using SHAP (SHapley Additive exPlanations). The SHAP values identified family unrest as the most important predictor with an importance of 61.8% (Figure 4). The next most impactful predictors were gender (36.5%), blackmail exposure (33.4%), internet addiction (29.7%), and pre-marital relationship (25.6%). Academic performance (SSC scores: 16.8%), major illness (14.9%), exercise habits (13.4%), and religious practice showed moderate importance.

LIME analysis of a representative test example (Figure 5) demonstrates how the features contribute to a single prediction. Family unrest contributed the most for high-risk classification. Blackmail exposure and gender were next. Lower levels of internet addiction ( $\leq -0.59$ ) and absence of major illness ( $\leq -0.39$ ) contributed negatively to risk classification, suggesting protective effects.

4) *Moderately Severe-or-Higher Depression Prediction (PHQ-9  $\geq 15$ )*: At the higher severity threshold (Threshold 4), the class distribution was more balanced (55.9% positive vs 44.1% negative); hence, SMOTE was not applied. Ada\_Log\_Stacking achieved the highest accuracy of 70.3%

TABLE V: Threshold 4 (PHQ-9  $\geq 15$ )

| Model                          | Acc.          | F1            | Prec.         | Rec.          | AUC           |
|--------------------------------|---------------|---------------|---------------|---------------|---------------|
| <i>Ensemble Model</i>          |               |               |               |               |               |
| Ada_Log_Stacking               | <b>0.7030</b> | <b>0.7409</b> | <b>0.7570</b> | 0.7257        | 0.7621        |
| <i>Machine Learning Models</i> |               |               |               |               |               |
| AdaBoost                       | 0.7025        | 0.7366        | 0.7492        | 0.7245        | <b>0.7662</b> |
| LogisticRegressionCV           | 0.7014        | 0.7373        | 0.7480        | 0.7270        | 0.7626        |
| LogisticRegression             | 0.7009        | 0.7370        | 0.7472        | 0.7272        | 0.7628        |
| LDA                            | 0.7004        | 0.7344        | 0.7482        | 0.7213        | 0.7627        |
| RidgeClassifier                | 0.6999        | 0.7341        | 0.7477        | 0.7213        | 0.7594        |
| LightGBM                       | 0.6863        | 0.7226        | 0.7262        | 0.7191        | 0.7287        |
| GradientBoosting               | 0.6842        | 0.7219        | 0.7220        | 0.7217        | 0.7481        |
| CatBoost                       | 0.6832        | 0.7234        | 0.7202        | 0.7267        | 0.7399        |
| <i>Deep Learning Models</i>    |               |               |               |               |               |
| ANN_DeepDrop                   | 0.6832        | 0.7145        | 0.7302        | 0.6997        | 0.7372        |
| ANN_Medium                     | 0.6702        | 0.7054        | 0.7165        | 0.6948        | 0.7244        |
| CNN_1D                         | 0.6702        | 0.7126        | 0.7085        | <b>0.7168</b> | 0.7198        |
| <i>Baseline</i>                |               |               |               |               |               |
| Dummy                          | 0.5586        | 0.7168        | 0.5586        | 1.0000        | 0.5000        |

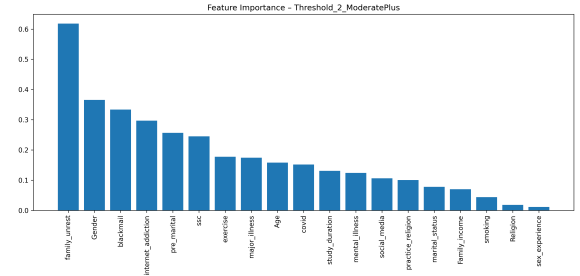


Fig. 4: Feature importance for PHQ-9  $\geq 10$  (Threshold 3) derived from L2-regularized Logistic Regression, showing family unrest as the dominant predictor.

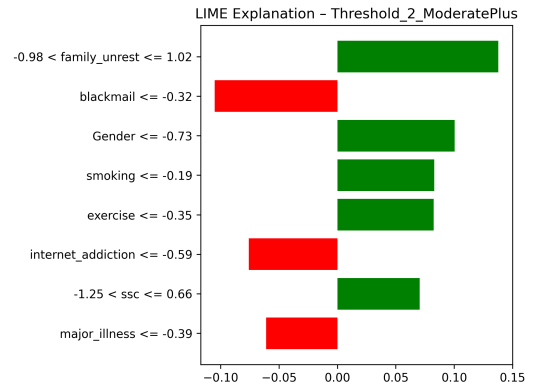


Fig. 5: LIME explanation for a representative prediction at PHQ-9  $\geq 10$  (Threshold 3). Green bars indicate features contributing positively to depression risk, while red bars indicate features reducing risk.

(F1 = 0.741, AUC = 0.762). Among individual models, AdaBoost performed best (70.3% accuracy, F1 = 0.737, AUC = 0.766), while deep learning (ANN\_DeepDrop: 68.3% accuracy, AUC = 0.737) and the Dummy classifier (55.9% accuracy, AUC = 0.500) showed moderate and baseline performance, respectively. Figure 6 presents ROC curves for top models (AUC  $\approx$  0.73–0.76). Ablation analysis indicated that AdaBoost (70.25%, AUC = 0.766) and LogisticRegressionCV (70.14%, AUC = 0.763) were the strongest base models, with Ada\_Log\_Stacking providing a small but consistent improvement (+0.05%) and enhancing robustness by combining nonlinear boosting with linear probabilistic modeling.

5) *Statistical Validation at Threshold 4: Friedman Test:* The Friedman test at Threshold 4 resulted in  $\chi^2 = 203.80$  with  $p < 0.001$  (Table VI), indicating highly significant performance differences among the 35 models. This suggests that algorithm selection remains important even at higher severity levels, although the top-performing models showed relatively similar performance.

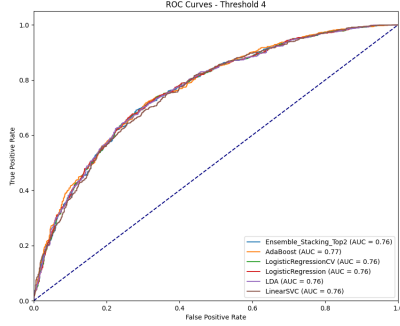


Fig. 6: ROC curves for top-performing models under PHQ-9  $\geq$  15 (Threshold 4).

6) *Feature Importance Analysis at Threshold 4:* L2-regularized Logistic Regression at Threshold 4 identified family unrest as the top predictor (55.6%), slightly lower than at Threshold 3 (Figure 7). Gender increased in relative importance (40.2%), followed by internet addiction (39.1%), SSC scores (26.7%), and pre-marital relationships (24.5%). Blackmail exposure decreased compared to Threshold 3 (22.4% vs 33.4%). Study duration (15.2%) and religious practice (14.9%) showed moderate contributions.

LIME analysis at Threshold 4 (Figure 8) shows differences in feature contributions compared to Threshold 3. Family unrest remains the top influencer on severe depression risk, while gender and religion (when  $\leq 0.26$ ) contribute positively. Internet addiction ( $\leq -0.59$ ) and blackmail exposure ( $\leq -0.32$ ) show negative contributions. These changes suggest that risk factors for severe depression differ from those for moderate depression.

## V. DISCUSSION

In this study, we used multi-threshold modeling, stacking ensemble learning, and statistical validation to evaluate de-

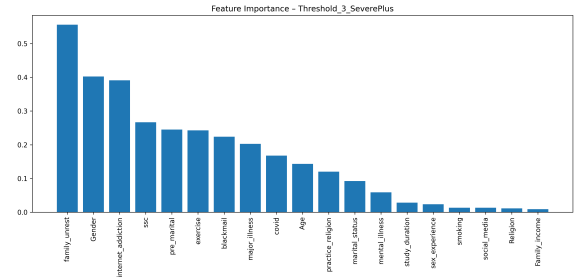


Fig. 7: Feature importance for PHQ-9  $\geq$  15 (Threshold 4) derived from L2-regularized Logistic Regression.

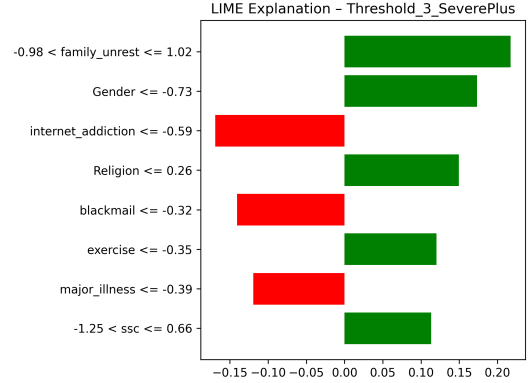


Fig. 8: LIME explanation for a representative prediction at PHQ-9  $\geq$  15 (Threshold 4). Green bars indicate features contributing positively to severe depression risk, while red bars indicate features reducing risk.

pression risk in Bangladeshi college-going adolescents. Compared to Siddik et al. (2024), which used traditional statistics [5], our approach focused on algorithm performance, feature importance, and threshold-specific risk patterns. The Friedman test confirmed significant differences among algorithms, and at the severe threshold, differences were highly significant. While simpler models like Logistic Regression and AdaBoost perform well in resource-limited settings, model choice still affects predictive reliability. Deep learning models performed moderately, likely due to sample size and limited features. Family unrest was the top predictor across thresholds, highlighting the role of family in adolescent mental health. Gender was the second most important predictor for severe depression (40.2%). These findings align with international studies [11], [12]. Family-focused screening and interventions are recommended, and for severe depression, simpler models like Logistic Regression and AdaBoost remain effective. Table VII summarizes key differences with prior research.

## Practical Deployment and Ethical Considerations

The model can serve as a screening-support tool rather than a clinical diagnostic system and could be integrated into school-based digital mental health programs. Misclassification risks remain: false positives may cause unnecessary concern,



TABLE VII: Comparison with prior research

| Aspect           | Current Study                  | Siddik (2024)        | Nayan (2022)     | Masud (2025) | Zhou (2024)     |
|------------------|--------------------------------|----------------------|------------------|--------------|-----------------|
| Population       | Adolescents                    | Adolescents          | University       | University   | Adolescents     |
| Sample Size      | 1,919                          | 1,919                | 2,121            | NR           | 8,635           |
| Methods          | 35 models + ensemble           | Statistics           | 5 models         | DL + SHAP    | ML models       |
| Thresholds       | Dual ( $\geq 10$ , $\geq 15$ ) | Single ( $\geq 10$ ) | Single           | Single       | Multiple        |
| Statistical Test | Friedman + Nemenyi             | None                 | None             | None         | NR              |
| Top Predictor    | Family unrest (56-62%)         | Family unrest        | Sociodemographic | Stress       | Family conflict |
| AUC              | 0.757-0.766                    | NR                   | NR               | NR           | 0.85            |
| Country          | Bangladesh                     | Bangladesh           | Bangladesh       | Bangladesh   | China           |

while false negatives may delay support. Predictions should therefore be interpreted as risk indicators and confirmed by mental health professionals. Strict data privacy, anonymization, informed consent, and non-punitive use are essential for ethical implementation.

#### Limitations and Future Work

Limitations include the cross-sectional design, self-reported measures, exclusion of out-of-college adolescents, moderate sample size for deep learning, potentially relevant predictors not included. This study was conducted using 10-fold cross validation to reduce overfitting ;however,the absence of external validation limits its generalizability. Although the multi-threshold classification models achieved consistent performance, the AUC values (approximately 0.75–0.76) indicate moderate discriminative ability rather than strong predictive certainty. Future work should use longitudinal designs, include diverse adolescent populations, incorporate richer clinical and behavioral data, test family-centered interventions.

#### VI. CONCLUSION

This study provides validated insights into adolescent depression in Bangladesh. We used multi-threshold modeling and stacking ensembles, which outperformed individual algorithms, showing that combining diverse models improves prediction in mental health datasets. 10-fold stratified cross-validation confirmed that models were not overfitting. Gradient boosting models performed best for moderate depression, while simpler linear models were better at higher severity levels. Family unrest was the strongest predictor across thresholds. By combining multi-threshold prediction, ensemble learning, and feature importance analysis, this study advances beyond prior single-threshold approaches and offers actionable insights for early detection and targeted intervention. We recommend family-centered interventions to improve communication and resolve conflicts. These results can guide mental health experts in better diagnosing depression among Bangladeshi adolescents.

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